Stanley Nelavala

Software Engineer, AI: Agents, Typed Tools and Evals (TypeScript, Python)

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Huntington, IN | Open to relocate anywhere

Summary

Software engineer with one year at a 10-person startup and an M.S. in Computer Science (4.0 GPA), who builds LLM agents people can trust: typed tools, human approval before anything consequential, 62 replay evals, prompt-injection tests and LLM failover. I measure before and after a change, and I code with Claude Code, Codex and opencode every day and check what they write with tests and acceptance checks. AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Built RxDesign, the hospital design service (FastAPI, React, TypeScript) with a deterministic, rule-based layout engine and an LLM setting; 220+ offline tests on the API run with fake models, so tests never call an LLM.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: row 150,000 of a 328,000-device dev database fell from 285 ms to 0.20 ms. Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search.
- Fixed a regular expression denial-of-service bug (138 s on an 80-character input); the new pattern matched the old results on 60,000 random inputs.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant; standardized malware scanning of uploads (GuardDuty) across 9 app modules.
- Use Claude Code, Codex and opencode every day, and help set the checks that keep agent output correct: tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis* GitHub

- Multi-step agent with typed state, one constrained MCP write tool, human approval that resumes after restart, and step and token budgets; hybrid retrieval with access labels.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction, an append-only audit log; failover between two LLM adapters with timeouts, a retry budget, a circuit breaker and token cost records.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch* GitHub

- Typed intent classifier service (int8, micro-batching, deadlines, load shedding): macro-F1 0.8636 for fp32 and int8 on the BANKING77 test split; one-factor experiments ruled out GC and tokenizer cost.

Payments Ledger Service | *Java, Spring Boot, PostgreSQL, React, TypeScript* GitHub

- Double-entry ledger with idempotency keys and a transactional outbox; property tests and a load check that no money is created or lost.

Skills

LLM agents: LangGraph, MCP tools, tool use, structured outputs, human-approval flows, deterministic evals and graders, guardrails, prompt-injection tests, PII redaction, LLM failover
AI coding agents: Claude Code and subagents, Codex, opencode, a DeepSeek harness; planner, manager and worker runs with acceptance checks
Languages: Python, TypeScript, JavaScript, SQL, Java, Go, Rust
Web and data: FastAPI, React, Node.js, PostgreSQL (pgvector, keyset paging), Redis, REST, gRPC
Cloud and quality: AWS (ECS, RDS, S3, SQS), Docker, GitHub Actions, pytest, Playwright, OpenTelemetry

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)